AWI-ESM3 ↔ PISM moving-cavity coupling

the investigation log

Single living document. Superseded claims are struck through, not deleted.

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Abstract

Where we are. PISM is in massive disequilibrium and is shedding its ice shelves: it loses **24% of its floating area per decade**, **1399 mesh nodes lose their ice entirely in a single 10-yr leg**, and the Ross front retreats ≈440 km. Our ocean **reacts violently** to that: it absorbs ∼99% of the change, but the remaining 1% (columns where the ice vanishes outright) is enough to kill the leg. **This is not good news.** Absorbing 99% is not a passing grade when the rest is fatal.

Two root causes we published and later falsified are struck through below. The current understanding: the cavity *geometry* is fine (a cold start on it is stable at the production timestep); the killer is the *seam* in our remapped restart, where patched values at the changed nodes meet a fully evolved dynamical state everywhere else.

Contents

1 Blocker I — OIFS step-0 coastline crash (RESOLVED)

A floating-point NaN struck exactly the coastline cells whose land–sea type flipped. `ocp-tool` half-converted a flipped cell (only mask + soil type), OIFS re-ingested the inconsistency from the regenerated `ICMGG` and NaN'd in moist physics. Fixed by a masked nearest-neighbour rebuild of each flipped cell's full column, plus the `ICMGG` re-ingest (`PUTALLFLDS_M` in `reresf_part2.F90`). A recurring `voskin` floating-overflow variant shows up state-dependently and was traced to the native-pipeline `ICMGG` coastal lsm.

2 Blocker II — FESOM cavity-margin blowup

2.1 Two root causes we got wrong

FALSIFIED #1 — "a geostrophically-unbalanced remapped state". ~~The mesh change plants fresh cavity fronts against salty open water but zeroes the balancing current; initialise the changed-cell velocity in thermal-wind balance instead of zero.~~ That fix was implemented (per-level $\mathbf{u}_g = (1/f\rho_0)\,\hat{z} \times \nabla p$, using FESOM's own cavity pressure convention) and **failed its own offline gate**: the discrete ∇p at grid-scale fronts is *noisier* than the nearest-neighbour fill. It ships opt-in only (`REMAP_GEOSTROPHIC=1`), off by default.

FALSIFIED #2 — "the sub-ice free-surface BC violation". ~~FESOM keeps $\eta \equiv 0$ under ice, yet the remap gave the 987 newly-sub-ice nodes their old open-ocean ssh (≈ -1.6 m); fixing that will cure the blowup.~~ A **real bug**, and fixed (esm_tools `b17fb5c8`) — but **not the cure**: the fixed run still died on day 4.4.

2.2 Every hypothesis, and the experiment that killed it

Hypothesis	Experiment	Verdict
Atmosphere / OASIS / coupling	ocean-only standalone FESOM reproduces the crash exactly (day 4, CFLz 9.24)	**ruled out**
CORE2 forcing shock (standalone artefact)	full-mesh control, *same* forcing + native restart: stable 21 d, worst CFLz **2.61**	**ruled out**
Remap corrupting the state	remapped vs. source over the 204 970 unchanged nodes: max $\lvert\Delta\rvert$ = **0.000** for `temp`, `salt`, `ssh`, `hnode`	**ruled out** (bit-exact)
The restart *fill values* at changed nodes	five independent fill strategies (donor, coherent-column, shelf-base hold, NN velocity, full balance hygiene)	**ruled out** (all die ≈day 4)
Pathological thin water columns	submesh min wet column 30 m vs. mother's 25 m; zero 1–2-layer columns	**ruled out** (submesh is *cleaner*)
Basal melt flux shock	`cavity_flux_ramp_days` implemented; melt effectively **OFF** still dies (mstep 328)	**ruled out**
Timestep / unresolved w	1200 s: CFLz kill day 4.0. 60 s: **0** CFLz warnings, dies day 4.4 via η_n. 120 s: dies day 8.9	**ruled out**
Sub-ice free-surface BC	real bug, fixed; run still dies day 4.4	**not the cure**
Cavity geometry itself	**cold start on the same PISM submesh @1200 s: STABLE, day 13+, worst CFLz 3.44**	**ruled out**

2.3 The decisive control: a cold start on PISM's cavity is stable

Same submesh, same forcing, same production timestep — only the initial state differs:

Run	Δt	Initial state	Outcome
full-mesh control	1200 s	native spun-up CORE3 restart	stable 21 d, worst CFLz **2.61**
PISM cavity, cold	**1200 s**	**cold (climatology, at rest)**	**stable 13+ d, worst CFLz 3.44**
PISM cavity, warm	1200 s	our remapped restart	**dies day 4.0**, CFLz 9.24
PISM cavity, warm	120 s	our remapped restart	dies day 8.9 (η_n)
PISM cavity, warm	60 s	our remapped restart	dies day 4.4 (η_n, 0 CFLz)

The ocean has no difficulty living in PISM's cavity at the production timestep. Whatever is wrong is in *our restart*, not in the mesh, the physics, or the coupling.

2.4 What is actually wrong: the dynamics seam

The remap produces an internally inconsistent state: on the **unchanged nodes (98%)** it reproduces the evolved restart **bit-exactly** (a fully developed, balanced dynamic state: $\mathbf{u}$, η, AB history); on the **changed nodes** it substitutes *patched* values. The two regions are mutually incompatible, and the **seam between them** is the instability. Zeroing the dynamics globally removes the crash (the ocean re-derives its own balanced flow from rest) — which is exactly why all five fill strategies failed: each varied the *patch* while leaving the *seam*. Zero-dynamics is not a production fix: discarding the ocean's circulation every leg is unacceptable.

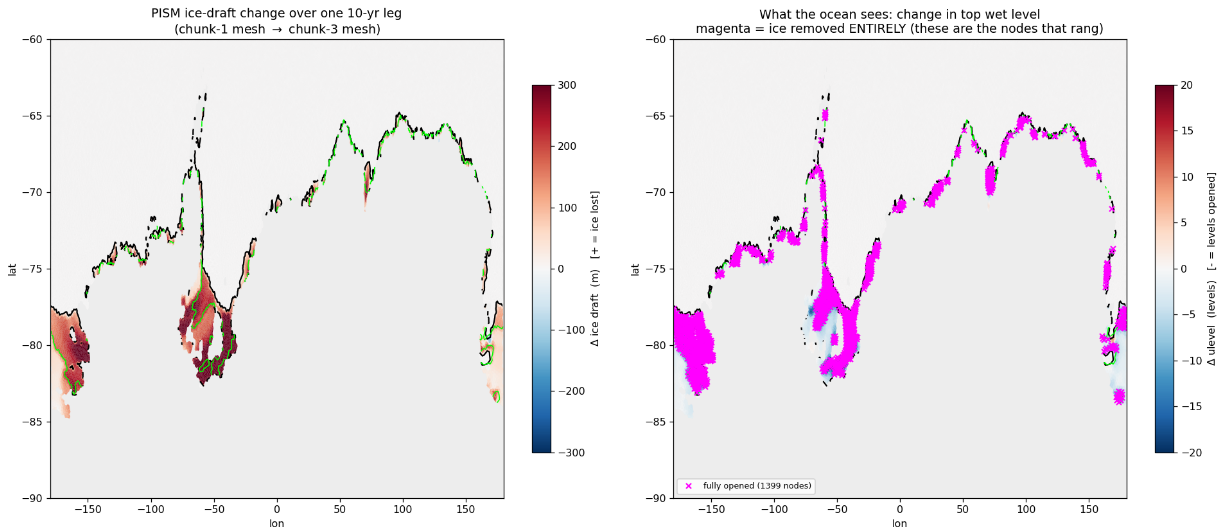
3 The real increment test (2026-07-13)

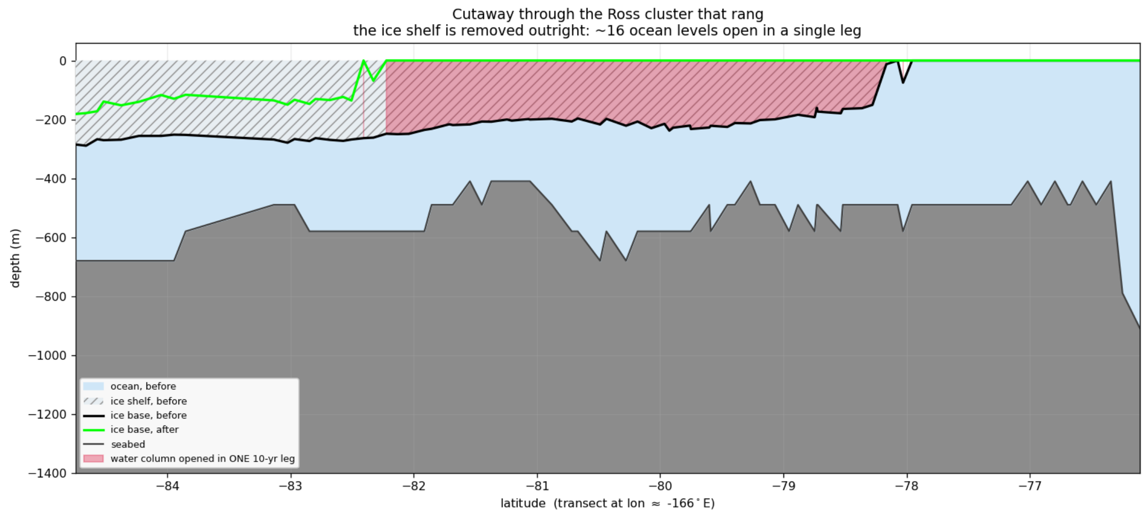
Cold-starting the ocean on PISM's cavity sidesteps the one-off CORE3(observed)→PISM swap ($\max|\Delta\text{draft}| = 2711$ m) entirely. That was done: chunk 1 ran a clean model year, PISM ran its 10 years, and the remap was finally asked to carry *a year of spun-up ocean dynamics across a real PISM increment.* It crashed on day 15.7.

	one-off swap (CORE3→PISM)	**real increment**
CFLz over time	4.46 → 6.43 → 7.25 → 9.24	5.26 → **5.78** → 5.40 → **4.15**
	monotonic runaway	**peaks, then decays**
died at	day ∼4	day 15.7
η_n NaN	yes	**none** — all 209 172 nodes finite
$\lvert\eta\rvert > 3$ nodes	3611, basin-wide	**17**, one cluster, **all changed nodes**

All 17 ringing nodes share one signature: `ulev 17 → 1` — **16 ocean levels opening at once**, i.e. the ice shelf above them removed *entirely.* Their η alternates in sign ($+24.4, -17.9, +15.9, +8.3$): a grid-scale **checkerboard**, not a basin mode.

3.1 The finding: PISM is collapsing





The cutaway is the clearest statement of the problem: the ice base goes from $-200\ldots-280$ m to **zero (open ocean)** across the whole span $-82.2°$ to $-78.2°$ latitude. **The Ross ice-shelf front retreats ≈440 km in ONE 10-yr leg.** The plan view shows this is not a Ross anomaly: the 1399 fully-opened nodes (magenta) ring the *entire* Antarctic margin.

changed nodes this leg	3248	(1.55% of mesh)
$\lvert\Delta\text{ulev}\rvert \geq 10$		1582
nodes with the ice removed ENTIRELY		**1399**
~~mean $\lvert\Delta\text{draft}\rvert$~~	~~2.9 m~~	← **meaningless**
max $\lvert\Delta\text{draft}\rvert$		**923 m**
PISM floating cells	24 815 → 20 093	(**−24% in 10 yr**)

FALSIFIED #3 — "the per-leg increment is gentle". ~~Production never has to survive the one-off swap; the per-leg PISM increment is two orders of magnitude smaller (mean |Δdraft| ≈ 22 m) and therefore absorbable.~~ **Wrong, and wrong in a way that mattered** — it motivated the entire "sidestep the swap and the increments will be fine" strategy. The mean is diluted by the 98% of the domain that does not change; the distribution is wildly skewed. PISM is *not* in equilibrium.

3.2 This is not good news

It is tempting to read the numbers optimistically: only 17 of 1399 fully-opened nodes rang; the ocean absorbed ∼99%; the CFLz transient peaked and *decayed*. All true, and it does mean the remap machinery is performing better than a "crashed at day 15" headline suggests.

But the honest reading is negative. The run still crashed. What we have learned is that **our ocean reacts violently to ice-shelf and cavity changes mid-run**: a handful of columns where the ice vanishes outright is enough to kill a leg. A coupled setup in which the ice sheet is *allowed* to change its cavity every 10 years cannot be one leg's worth of extreme nodes away from failure. **Absorbing 99% of the change is not a passing grade when the remaining 1% is fatal.**

4 Where to go from here

1. **Find out why PISM is collapsing, and stop it.** A run shedding 24% of its floating area in a decade is the anomaly, and everything downstream inherits it. **The first thing to check is the sub-shelf melt *we* feed PISM**: if our ocean hands it unrealistic melt, then PISM's collapse is our own doing, and the loop closes.

2. **Try to stabilise the ocean locally.** Reduce the ocean timestep and/or raise viscosity *in the neighbourhood of the changed columns*, and see whether the transient can be damped enough to survive. A mitigation, not a cure — but it may keep the coupling running while (1) is sorted out. Note the *global* knobs are already exhausted: a global timestep cut does **not** help (60 s merely swaps the CFLz symptom for an η_n one).

3. **The lasting way forward: compute physically consistent fields for the newly opened cavity.** Patching such a column with plausible *values* is what we have been doing, and it is exactly what does not work (five fill strategies, all dead; §2.4). What is needed is a state for the newly opened water that is **dynamically consistent with its surroundings**, not merely a sensible number in each cell. That is the real problem, and the one worth solving.

A per-leg cap on $|\Delta\text{ulev}|$ (defer the extreme tail across legs) remains available as a blunt robustness guard, but it papers over (1) rather than fixing anything.

5 Bugs found and fixed along the way

5.1 In the remap

Unbounded extrapolation below the old seabed (esm_tools `eef4f6db`). The first attempt at the increment leg died at **mstep 1**: the remapped restart contained −45.02°C, a value present *nowhere* in the source (wet range [−2.32, 41.72], dry cells 0.0). The remap *manufactured* it. The column shows the mechanism: 1.22, −0.99, −3.19, −5.39, −8.69, −13.09, −18.6, −25.2, −34.01, **−45.02** — smooth and accelerating. For columns that get *deeper*, the remap fitted a straight line through the last two old levels and continued it downward with no clamp. Fixed by taking such water from the nearest old node actually wet at that depth. **Never hit before** because every previous remap went mother → submesh (a strict *subset*); ice *retreating* grows the mesh and exercises this path for the first time.

Also: sub-ice `ssh`/`hbar` zeroing (`b17fb5c8`); a **size guard** so the remap refuses a restart that does not match the old mesh, instead of silently writing garbage (`3b44a704`).

5.2 In the coupling plumbing

Making chunk 1 a *submesh* leg pushed a submesh leg through `couple_out` for the first time. Every previous run either ran chunk 1 on the *full mesh* (where these paths are trivially correct) or died early in chunk 3 (before reaching the end of a leg). That exposed:

- **OASIS `GSSPOS` abort at ∼day 300.** Its conservation correction is built from the summed source/target ratio; when the summed target residual is ≈ 0 while the source residual is small-but-nonzero, it explodes. Only the heat-to-ice flux (`A_Q_ice` → `heat_ico`) used it. Fixed: new `gauswgt_gsmart` transform (`3e94d4da`).
- **The namcouple `feom` dimension had NEVER been patched, on any leg.** `fix_namcouple_feom_dim` read `${MAXMESH_DIR_fesom}`, which is assigned *inside* `build_submesh` — a different subjob's shell. Always unset ⇒ guard refused ⇒ silent no-op. Fixed (`3cf29e9e`).
- `fesom2ice` **loaded the static full mesh** while FESOM's output is on the submesh (broadcast error) (`63cfcc91`).
- `previous_submesh` **is read but never created** — anywhere. Every leg from chunk 5 on would have remapped against the wrong mesh (`3b44a704`).
- **A failed `couple_in` did not stop the workflow.** The coupling function is backgrounded, the observe job is not its parent (so it can never see the exit code), and `observe.py` returns an *empty* error list for every non-compute job. The next leg started on an un-remapped restart. Fixed with a `.coupling_failed` sentinel (`3b44a704`).

The pattern: the guards were right, and the plumbing around them defeated them. `remap_oasis_restart` correctly refused a mis-sized restart — and the run proceeded anyway.

6 Assets

- **A 2-minute offline reproducer**: standalone ocean-only FESOM on the PISM-cavity submesh, 14 nodes, crashes in ∼2 min (vs. the 45-node coupled system). Requires the `FESOM_COUPLED=OFF` build at the current SHA.
- **A stable control**: full mesh + native restart + same forcing (21 d, CFLz 2.61).
- **Chunk-1 artifacts pooled** at `input/fesom2/pism_cavity_ini/` (540 MB), so chunk 1 needs no `couple_in` and `esm_runscripts` can be run straight from the login node.
- `cavity_flux_ramp_days` (FESOM `&run_config`, default 0 = off, bit-identical).
- `verify_balance.py`: offline balance/stability gate (min N^2, max $|\nabla\cdot(H\bar{\mathbf{u}})|$, CFL, Ri) — it caught the geostrophic ∇p init being worse than the NN baseline *before* it cost a model run.

7 Method note

Three root causes were asserted with more confidence than the evidence supported, and all three were later falsified by a control that was cheap and should have come *first*:

1. "geostrophic imbalance" — killed by the **cold start** (is it the geometry or our state?);
2. "the sub-ice ssh BC" — killed by simply **running the fix**;

3. “the increment is gentle (≈22 m)” — killed by **plotting the distribution** instead of trusting a mean over a domain that is 98% unchanged.

The third is the sharpest: the 22 m figure was computed correctly and reported honestly, and it was still misleading enough to steer the strategy for a day. **Two figures did in ten minutes what a summary statistic obscured for a day.**